LAB REPORT 05
Discrete-Time Systems & Laplace Transforms

Exercise 01: Discrete-Time System Frequency & System Response

> Suppose that a discrete-time system is described by the frequency response:
> $H(\omega) = (4 + 2e^{(-2j\omega)}) / (3 + 5e^{(-j\omega)})$
>
> Plot the frequency response $H(\omega)$. Also, compute and plot the system response to the discrete-time input signal:
> $x[n] = 2\cos(4n + \pi/6)$
>
> Numerical Outputs:
> • $H(4)$: -0.589430 - 0.938397j
> • Magnitude $|H(4)|$: 1.108159
> • Phase $\angle H(4)$: -2.131640 radians
> • Output Amplitude (A): 2.216318
> • Output Phase (ϕ): -1.608041 radians

MATLAB Implementation Code:

```
clc;
clear;
close all;

%% Part 1: Frequency Response H(w)
% Define frequency range
w = linspace(-pi, pi, 2000);

% Define system frequency response
H = (4 + 2*exp(-1j*2*w)) ./ (3 + 5*exp(-1j*w));

% Plot Magnitude and Phase Response
figure(1);

subplot(2,1,1);
plot(w, abs(H), 'b', 'LineWidth', 1.5);
grid on;
xlabel('Frequency (rad/sample)');
ylabel('|H(w)|');
title('Magnitude Response');

subplot(2,1,2);
plot(w, unwrap(angle(H)), 'r', 'LineWidth', 1.5);
grid on;
xlabel('Frequency (rad/sample)');
ylabel('Phase (radians)');
title('Phase Response');
```

```
%% Part 2: System Response to Input x[n]
% Time index and input signal parameters
n = 0:40;
x = 2 * cos(4*n + pi/6);
w0 = 4;

% System response at frequency w0 = 4
H0 = (4 + 2*exp(-1j*2*w0)) / (3 + 5*exp(-1j*w0));

% Output magnitude and phase
A = 2 * abs(H0);
phi = pi/6 + angle(H0);

% Compute output signal y[n]
y = A * cos(w0*n + phi);

% Display numerical results
fprintf('H(4) = %.6f + %.6fj\n', real(H0), imag(H0));
fprintf('Magnitude of H(4) = %.6f\n', abs(H0));
fprintf('Phase of H(4) = %.6f radians\n', angle(H0));
fprintf('Output amplitude = %.6f\n', A);
fprintf('Output phase = %.6f radians\n', phi);

% Plot Input and Output signals
figure(2);

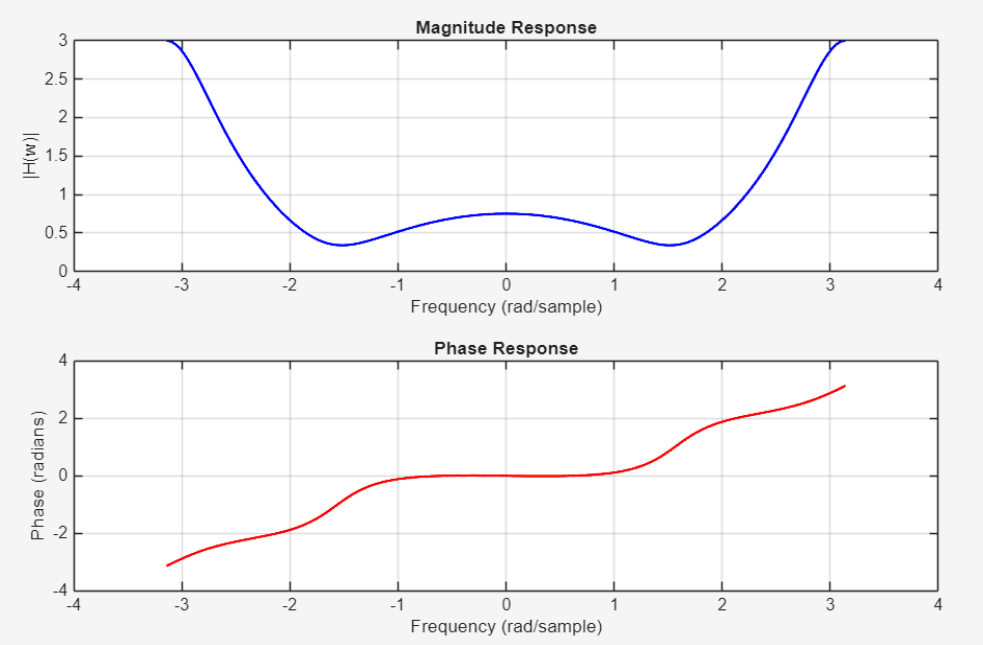
subplot(2,1,1);
stem(n, x, 'filled', 'b');
grid on;
xlabel('n');
ylabel('x[n]');
title('Input Signal x[n]');

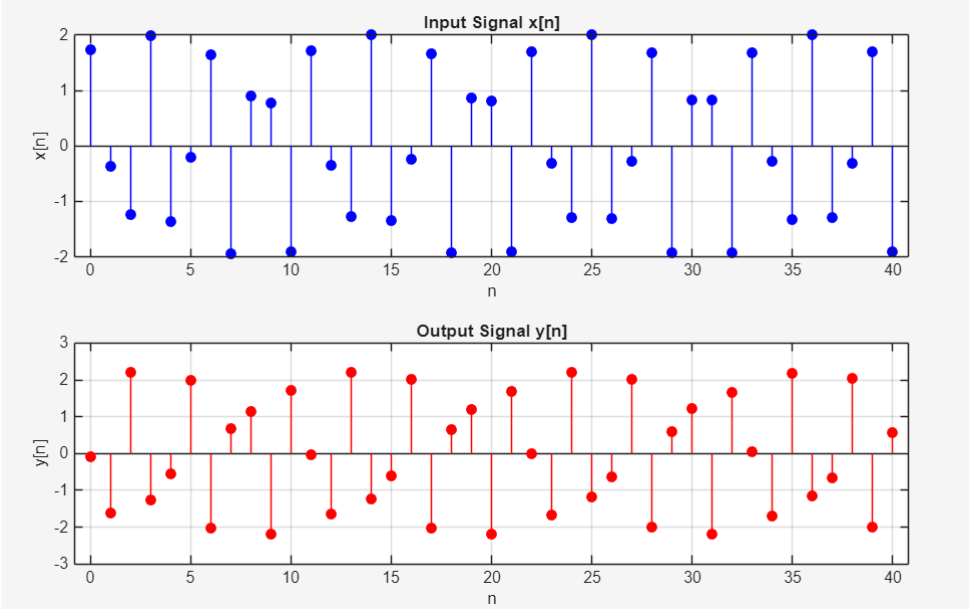
subplot(2,1,2);
stem(n, y, 'filled', 'r');
grid on;
xlabel('n');
ylabel('y[n]');
title('Output Signal y[n]');
```

Simulation Output Plot:

Command Window

```
H(4) = -0.589430 + -0.938397j
Magnitude of H(4) = 1.108159
Phase of H(4) = -2.131640 radians
Output amplitude = 2.216318
Output phase = -1.608041 radians
>>
```





Exercise 02: Unilateral Laplace Transform

Compute the unilateral Laplace transform of the following functions:
- 2(a): δ(t)
- 2(b): u(t)
- 2(c): e^(-3t)u(t)
- 2(d): cos(2πt)u(t)

Analytical Results:
• 2(a) $\mathcal{L}\{\delta(t)\}$: 1
• 2(b) $\mathcal{L}\{u(t)\}$: 1 / s
• 2(c) $\mathcal{L}\{e^{-3t}u(t)\}$: 1 / (s+3)
• 2(d) $\mathcal{L}\{\cos(2\pi t)u(t)\}$: $s / (s^2 + 4\pi^2)$

MATLAB Implementation Code:

```
clc;
clear;
close all;

% Define symbolic variables
syms t s

% 2(a): Laplace Transform of Dirac delta function
f1 = dirac(t);
F1 = laplace(f1, t, s);

% 2(b): Laplace Transform of Heaviside step function u(t)
f2 = heaviside(t);
F2 = laplace(f2, t, s);

% 2(c): Laplace Transform of exp(-3t)*u(t)
f3 = exp(-3*t) * heaviside(t);
F3 = laplace(f3, t, s);

% 2(d): Laplace Transform of cos(2*pi*t)*u(t)
f4 = cos(2*pi*t) * heaviside(t);
F4 = laplace(f4, t, s);

% Display results
disp('2(a):'); disp(F1);
disp('2(b):'); disp(F2);
disp('2(c):'); disp(F3);
disp('2(d):'); disp(F4);
```

Simulation Output Plot:

Command Window

```
2(a):
1

2(b):
1/s

2(c):
1/(s + 3)

2(d):
s/(4*pi^2 + s^2)

>>
```